HaqSetu — AI Welfare and Scholarship Navigator (Lean Canvas)

Primary SDG: SDG 4 (Quality Education) | Secondary: SDG 1 (No Poverty), SDG 10 (Reduced Inequalities) | IBM SkillsBuild Internship Deliverable

1. Problem

• Eligible students miss scholarships due to fragmented PDFs and unclear criteria.
• Generic chatbots hallucinate income limits, marks cutoffs, and deadlines.
• Students lack a clear prerequisite order for certificates and OTR registration.

4. Solution (Implemented MVP)

• Deterministic 3-valued Kleene eligibility engine (ELIGIBLE / INELIGIBLE / NEEDS_VERIFICATION).
• DAG Action Planner (graphlib.TopologicalSorter) with DONE/READY/BLOCKED/GATED states.
• Stdlib TF-IDF policy clause retriever with strict refusal when evidence is unverified.

3. Unique Value Proposition

• Eligibility Condition -> Supporting Evidence -> Decision -> Next Action.
• Zero hallucinated verdicts: unverified schemes fail closed to LISTING_ONLY.
• Every evaluated condition links to its exact clause, section, and source URL.

9. Unfair Advantage

• Strict structural separation between SchemeReadiness (ASSESSABLE vs LISTING_ONLY) and ApplicantOutcome.
• Safety gates cannot be bypassed by marking earlier steps complete.

2. Customer Segments

• Primary: Indian school and college students seeking public scholarships (SDG 4/1/10).
• Secondary (Hypothesis): School/college scholarship cells, CSCs, NGO facilitators.

8. Key Metrics (Actual Implemented)

• Automated pytest suite: 32 unit & integration tests across engine, planner, retriever, API.
• Retrieval benchmark (10 synthetic queries): Top-1=1.0, Top-3=1.0, MRR=1.0, Refusal=1.0 (small synthetic benchmark — not independent validation).
• Zero external ML/LLM dependencies; runs locally on Python 3.14.

5. Channels (Hypothesis)

• Local offline-capable institutional kiosk deployment.
• Open-source educational utility for college nodal cells and NGO field workers.

7. Cost Structure (Actual MVP)

• Zero cloud hosting, database, or paid LLM API costs.
• Runs locally on standard student laptop (Python 3.14, FastAPI, Pydantic v2, vanilla JS).

6. Sustainability / Revenue (Hypothesis)

• Public-good open-source tool aligned with SDG 4.
• Potential CSR/NGO grant sponsorship for human policy verification cycles (Hypothesis).

Honest Scope & Verification Note

• Real candidates (PM-USP, UP Pre-Matric) remain LISTING_ONLY until human verification of current cycle guidelines.
• Interactive assessment demonstrated safely via explicitly labelled FICTIONAL_FIXTURE.
• No RAM footprint claim published — not measured in an automated test.